S-BLOCK

ALKALI METALS

1) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$ ~~$< Fr^+$~~ (Size) (Gaseous) (hydrated mein exactly opposite)

2) $Li < Na < K < Rb < Cs$ (Size)

3) $Li > Na > K > Rb > Cs$ (IE)

4) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ (HE)

★ Cs for photoelectric effect

5)

Metal	Flame Colour
Li	Crimson Red
Na	Yellow
K	Violet
Rb	Red Violet
Cs	Blue

6) $Li > Na > K > Rb > Cs$ (M.p)

7) $Li > Na > K > Rb > Cs$ (B.p)

8) $Li < Na > K < Rb < Cs$ (Density)

(0.53) (0.97) (0.86) (1.53) (1.90)

9) $Li > K \simeq Rb \simeq Cs > Na$ (SRP) / (~~Oxidising~~ Reducing strength)

10) Reacⁿs →

a) $4Li + O_2 \rightarrow 2Li_2O$

$2Na + O_2 \rightarrow Na_2O_2$

$M + O_2 \longrightarrow MO_2$ ($M = K, Rb, Cs$)

$6Li + N_2 \longrightarrow 2Li_3N$

b) $2M + 2H_2O \rightarrow 2M^+ + 2\bar{O}H + H_2$ (Li less vigrous than Na)

c) $2M + H_2 \longrightarrow 2M^+H^-$

d) $M^+ + X^- \longrightarrow MX$

Li halides covalent (high polarisation)

e) $M + (x+y)NH_3 \longrightarrow [M(NH_3)_x]^+ + [e(NH_3)_y]^-$ → R.A also

Blue colour due to ammoniated e^-

Paramagnetic

[Impurity Present] → $M^+ + e^- + NH_3 \longrightarrow MNH_2 + \boxed{\frac{1}{2}H_2}$ → liberate H_2

Conc^n soln → Blue changes to bronze and diamagnetic

(forma. of metal ion clusters)

11) Uses →

Li → White metal

Na → Na/Pb alloy to make $PbEt_4$ and $PbMe_4$

12) Reac^{ns} →

$M_2O + H_2O \longrightarrow 2M^+ + 2\bar{O}H$

$M_2O_2 + 2H_2O \longrightarrow 2M^+ + 2\bar{O}H + \boxed{H_2O_2}$

$2MO_2 + 2H_2O \longrightarrow 2M^+ + 2\bar{O}H + \boxed{H_2O_2 + O_2}$

↓

yellow/orange

Superoxides → Paramagnetic

13) $LiF < NaF < KF < RbF < CsF$ ($\Delta H_f^\ominus$ values)

less -ve

14) $LiX > NaX > KX > RbX > CsX$ ($\Delta H_f^\ominus$ values)

More -ve

15) fluoride > chloride > bromide > Iodide (Mp, bp order with Li, Na, K, Rb, Cs

16) LiF, CsI were less soluble

17) LiCl soluble in pyridine also

18) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

$X_2CO_3 \xrightarrow{\Delta} XX$ (X = Na, K, Rb, Cs)

$XHCO_3 \xrightarrow{\Delta} {}^{M_2CO_3}_{+H_2O}{}_{+CO_2}$ (X = ~~Li~~, K, Rb, Cs, Na), $LiHCO_3 \xrightarrow{\Delta} Li_2O + H_2O + 2CO_2$

19) Li diagnol relationship with Mg

20) i) Li is harder than K, Rb, Cs, Na

ii) LiCl → deliquescent

$LiCl + H_2O \rightarrow LiCl.2H_2O$

iii) $Li + HC \equiv CH \rightarrow XX$

iv) $4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$] → Same with K, Rb, Cs

v) $LiHCO_3$ is not obtained in solid form whereas others in solid

21) Li < Mg , $Li^+ > Mg^{2+}$ (Size)

22) Li_2O & $MgO + O_2$ (excess) $\longrightarrow XX$

23) LiCl, $MgCl_2$ → soluble in ethanol

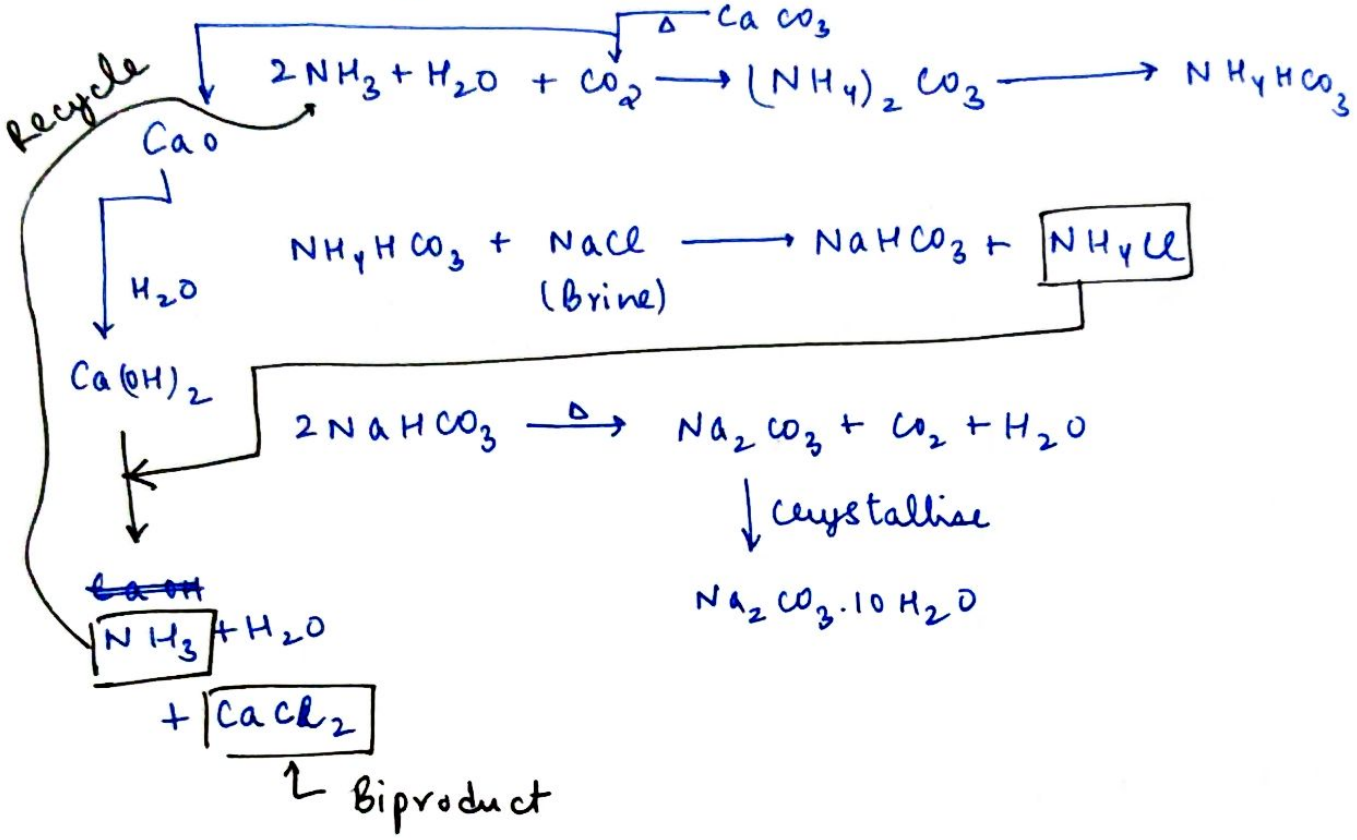
24) $LiCl + H_2O \rightarrow LiCl.2H_2O$

$MgCl_2 + H_2O \rightarrow MgCl_2.8H_2O$

25) $Na_2CO_3 \cdot 10H_2O$ (Washing Soda)

a) ↳ ① Prepared by Solvay Process

② $NaHCO_3$ ⇌ is least soluble



For $KHCO_3$, solvay xx as more soluble

b) $Na_2CO_3 \cdot 10H_2O \xrightarrow{315K} Na_2CO_3 \cdot H_2O + 9H_2O$

$Na_2CO_3 \cdot H_2O \xrightarrow{>373K} Na_2CO_3 + H_2O$

C) Le Blanc Process

$NaCl + \text{conc. } H_2SO_4 \longrightarrow NaHSO_4 + HCl$

↓ NaCl

$Na_2SO_4 + HCl$

'C' ↓ $CaCO_3$

$\underline{Na_2CO_3 + CaS + CO}$

↳ Black Ash

26) NaCl

↳ evaporation of Sea water

27) NaOH (Caustic Soda)

(a) Electrolysis → Nelson / Diaphragm Cell

i) $H_2O + e^- \rightarrow \frac{1}{2} H_2 + HO^-$

ii) $Cl^- - e^- \rightarrow \frac{1}{2} Cl_2$

iii) NaOH

(b) Castner Kellner Cell →

Anode → Cl_2

Cathode → Hg coated

↳ $Na^+ + e^- \rightarrow Na \xrightarrow{Hg} Na_2Hg_x$

$\downarrow 2H_2O$

$2NaOH + xHg + H_2\uparrow$

28) Baking Soda ($NaHCO_3$)

$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$

↓

antiseptic

27) Conti →

(c) Gossage Process

$Na_2CO_3 + Ca(OH)_2 \longrightarrow CaCO_3\downarrow + 2NaOH$

(d) Lewig Process →

$Na_2CO_3 + Fe_2O_3 \longrightarrow NaFeO_2 + CO_2$

$NaFeO_2 + H_2O \longrightarrow Fe_2O_3 + 2NaOH$

ALKALINE EARTH METALS →

1) $Be < Mg < Ca < Sr < Ba$

2) $Be^{+2} < Mg^{2+} < Ca^{+2} < Sr^{+2} < Ba^{+2}$ [Size]

3) $Be > Mg > Ca > Sr > Ba > Ra$ (IE_1)

4) $Be > Mg > Ca > Sr \gtrless Ba < Ra$ (IE_2)

5) $Be > Mg > Ca > Sr > Ba$ (HE) → Magnitude

← More -ve

6) $Be > Ca > Sr > Ba > Mg$ (mp)

7) $Be > Ba > Ca > Sr > Mg$ (bp)

8) $Ba > Sr > Be > Mg > Ca$ (Density)

9) $Ba > Sr > Ca > Mg > Be$ (SRP (M^{+2}/M))

→ More -ve

10)

Flame Colour	Metal
X	Be] → e⁻ not excited / visible region mein no excitation
X	Mg]
Brick Red	Ca
Crimson Red	Sr
Apple Green	Ba

11) $Be/Mg + O_2 \longrightarrow XX$ (oxide film)

Powdered $Be/Mg \xrightarrow{O_2} BeO/MgO$

$\xrightarrow{N_2} Be_3N_2 / Mg_3N_2$

12) $M + X_2 \longrightarrow MX_2$

$(NH_4)_2BeF_4$ best for Prep. of BeF_2

13) $BeO + C + Cl_2 \underset{}{\overset{600-800K}{\rightleftharpoons}} BeCl_2 + CO$

14) $2BeCl_2 + LiAlH_4 \longrightarrow 2BeH_2 + LiCl + AlCl_3$

15) $M + (x+y)NH_3 \longrightarrow [M(NH_3)_x]^{2+} + 2[e(NH_3)_y]^-$

$[M(NH_3)_6]^{2+}$ can be recovered

16) Milk of Magnesia ($Mg(OH)_2$) is used as antacid

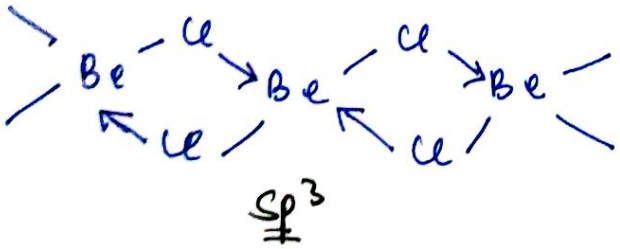
17) $MO + H_2O \longrightarrow M(OH)_2$

$Be(OH)_2 + 2\bar{O}H \longrightarrow [Be(OH)_4]^{2-}$

(Beryllate ion)

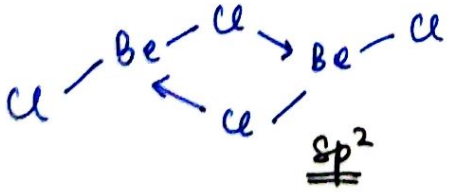
$Be(OH)_2 + 2HCl + 2H_2O \longrightarrow [Be(OH)_4]Cl_2$

18) $BeCl_2 \rightarrow$



sp^3

Vapour Phase →



sp^2

↓ T↑ above 1200K

Cl – Be – Cl

sp

19) Mg > Ca > Sr > Ba (Tendency to form hydrates)

Mg → $.8H_2O$
Ca → $.6H_2O$
Sr → $.6H_2O$
Ba → $.2H_2O$

20) All carbonates $\xrightarrow{\Delta}$ $CO_2 + H_2O$

$BeCO_3$ only kept in CO_2 atmosphere [$MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$ (stab. towards Δ)]

21) Sulphates stable to heat

$BeSO_4 \simeq MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$ (solubility in water)

22) $2M(NO_3)_2 \longrightarrow 2MO + 4NO_2 + O_2$

↓

Be | Mg | Ca | Sr | Ba

23) Be | Al diagnol relationship

$BeCl_2$, $AlCl_3$ → Freidal Craft Catalysts

BeF_4^{2-}, AlF_6^{3-}

↑ strong tendency to form this

24) Quick lime CaO

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

$CaO + H_2O \longrightarrow Ca(OH)_2$ (slaking of lime)

$CaO + CO_2 \longrightarrow CaCO_3$

Sodalime → $NaOH + CaO$

$CaO + SiO_2 \longrightarrow CaSiO_3$

$CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2$

95) Slaked lime ($Ca(OH)_2$)

$$CaO + H_2O \rightarrow Ca(OH)_2 \xrightarrow{CO_2} \underset{\text{Milky}}{CaCO_3} + H_2O$$

$$\downarrow H_2O, CO_2$$

$$Ca(HCO_3)_2 \rightarrow \text{Milkiness} \times\times$$

$$2Ca(OH)_2 + 2Cl_2 \rightarrow CaCl_2 + Ca(OCl)_2 + 2H_2O$$

(Bleaching Powder)

$$CaCO_3 \xrightarrow{1200K} CaO + CO_2$$

96) POP $\rightarrow CaSO_4 \cdot \frac{1}{2} H_2O$

Gypsum $\rightarrow CaSO_4 \cdot 2H_2O$

$CaSO_4 \rightarrow$ dead burnt Plaster

Advance Level Points →

1) $CsOH > RbOH > KOH > NaOH > LiOH$ (Solubility)

2) $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$ (Solubility)

3) $Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$ (Stabi.)

4) $KO_2 < RbO_2 < CsO_2$ (Stabi.)

5) Reactivity with O_2 → $Li < Na < K < Rb < Cs$

$$Na + O_2 \text{ (excess)} \longrightarrow Na_2O_2 \xrightarrow{H_2SO_4} H_2O_2 + Na_2SO_4$$

$$Na_2O_2 \xrightarrow[\Delta]{H_2SO_4} H_2O + O_2 + Na_2SO_4$$

6) Na, oxyg. comp. → Red litmus blue → Na_2O

Red litmus white → Na_2O_2 (Bleaching)

7) $12NaOH + S_8 \longrightarrow 2Na_2S_2O_3 + 4Na_2S + 6H_2O$

$$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$$

$Cl_2 / Br_2 / I_2$

- Cold, dil NaOH → $NaOCl + NaCl + H_2O$ (Hypochlorite)
- Conc., Hot NaOH → $NaClO_3 + NaCl + H_2O$ (Chorate)

8) NaOH

BeO → Na_2BeO_2

ZnO → Na_2ZnO_2

PbO → Na_2PbO_2

Al_2O_3 → $NaAlO_2$

SnO → Na_2SnO_2

PbO_2 → Na_2PbO_3

SnO_2 → Na_2SnO_3

$NH_4Cl + \Delta$ → $NH_3 + H_2O + NaCl$

Reactant is amphoteric

$(NH_4)_2SO_4$ → $NH_3 + H_2O + Na_2SO_4$

9) NaOH (fused)

Zn → $Na_2ZnO_2 + H_2$

Sn → $H_2 + Na_2SnO_2$

Al → $H_2 + NaAlO_2$ (used to open drains)

10) $MgCl_2 \cdot 6H_2O$

Δ → $Mg(OH)_2 + HCl + H_2O$

→ $MgCl_2$

Heat in current / steam of HCl

11) NaOH

$SnCl_2$ → $Sn(OH)_2$ → NaOH → Soluble

$ZnCl_2$ → $Zn(OH)_2$ → NaOH → Soluble

$AlCl_3$ → $Al(OH)_3$ → NaOH → Soluble

$FeCl_3$ → $Fe(OH)_3$ → NaOH → not Soluble

$CrCl_3$ → $Cr(OH)_3$ → NaOH → Partially Soluble

12) $Ca(CH_3COO)_2$ and $Ce_2(SO_4)_3$ → Solubility with T ↑

↳ dissolu. is exo

13) $2NaOH + 4NO \longrightarrow 2NaNO_2 + N_2O + H_2O$ (2:4)

$4NaOH + 6NO$ (excess) $\longrightarrow 4NaNO_2 + N_2 + 2H_2O$ (4:6)

$KI + H_2SO_4 + NaNO_2 \longrightarrow NO + I_2 + K_2SO_4 + Na_2SO_4 + H_2O$

14) $Na_2O_2 + H_3PO_4$ gives H_2O_2

15) $CO + NaOH \xrightarrow[P\uparrow]{T\uparrow} HCOONa \xrightarrow{\Delta} Na_2C_2O_4 \xrightarrow{H^+} H_2C_2O_4$

$Na_2C_2O_4 \xrightarrow{CaCl_2} Ca_2C_2O_4$ (white)

16) $Cs_2O > K_2O > SrO > MgO$

$Na_2O > SrO$ [Basic Charac.)

17) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$ (Conductance)

18) $Cs > Rb > K > Na > Li$ (Reactivity with H_2O)

$Li > Na > K > Rb > Cs$ (Reac. with H_2O)

$LiH > NaH > KH > RbH > CsH$ (stabi.

$LiOH < NaOH < KOH < RbOH < CsOH$ (Thermal stab.

19) $O_3 \xrightarrow[dry]{KOH} KO_3 + O_2 + H_2O$

(KO_3 (Pota. Ozonide) (Reddish orange Colour. Comp.)

20) $KO_2 + S \longrightarrow K_2SO_4 \xrightarrow[(aq)]{Al_2(SO_4)_3} K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(Potash Alum)

21) Read hard / soft water from notes